



(12) **United States Patent**
Jeon

(10) **Patent No.:** **US 12,412,636 B2**
(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **STORAGE DEVICE, HOST DEVICE, AND METHOD OF OPERATING THE SAME**

(71) Applicant: **SK hynix Inc.**, Gyeonggi-do (KR)

(72) Inventor: **Jeong Ho Jeon**, Gyeonggi-do (KR)

(73) Assignee: **SK hynix Inc.**, Gyeonggi-do (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/748,097**

(22) Filed: **Jun. 20, 2024**

(65) **Prior Publication Data**

US 2024/0339167 A1 Oct. 10, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/748,421, filed on May 19, 2022, now Pat. No. 12,027,222.

(30) **Foreign Application Priority Data**

Dec. 8, 2021 (KR) 10-2021-0175075

(51) **Int. Cl.**
G11C 29/08 (2006.01)
G11C 16/04 (2006.01)
G11C 16/08 (2006.01)

(52) **U.S. Cl.**
CPC **G11C 29/08** (2013.01); **G11C 16/0483** (2013.01); **G11C 16/08** (2013.01)

(58) **Field of Classification Search**

CPC G11C 29/08; G11C 16/0483; G11C 16/08; G11C 29/04; G11C 29/46; G11C 16/26; G11C 2029/0409; G11C 2029/0411; G11C 29/42; G11C 29/56; G06F 3/061; G06F 3/0659; G06F 11/1048; G06F 12/0246

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

12,027,222 B2* 7/2024 Jeon G11C 29/42
2019/0051359 A1* 2/2019 Yoo G11C 16/10

FOREIGN PATENT DOCUMENTS

KR 10-2019-0018323 A 2/2019

OTHER PUBLICATIONS

Office Action for Korean Patent Application No. 10-2021-0175075 issued by the Korean Patent Office on May 29, 2025.

* cited by examiner

Primary Examiner — Albert Decady

Assistant Examiner — Osman M Alshack

(74) *Attorney, Agent, or Firm* — IP & T GROUP LLP

(57) **ABSTRACT**

A method of operating a host device according to the present technology includes determining an area to be tested among a mapped area and an unmapped area included in a storage area of a storage device, generating a test request corresponding to the determined area, and transmitting the generated test request to the storage device.

13 Claims, 18 Drawing Sheets

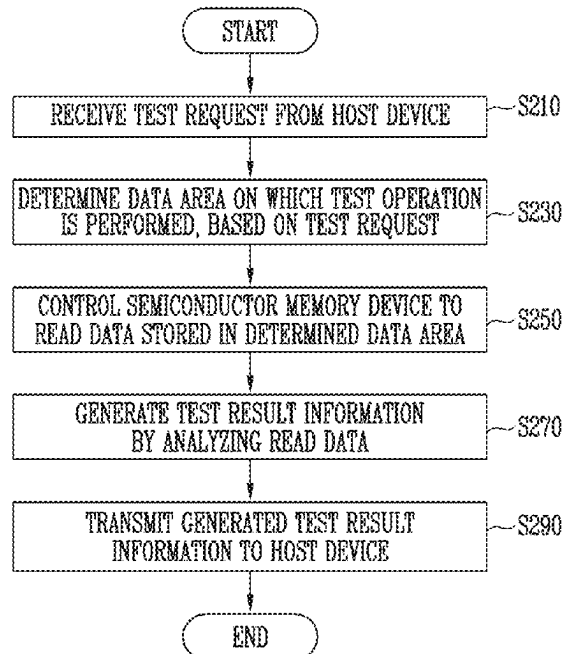


FIG. 1

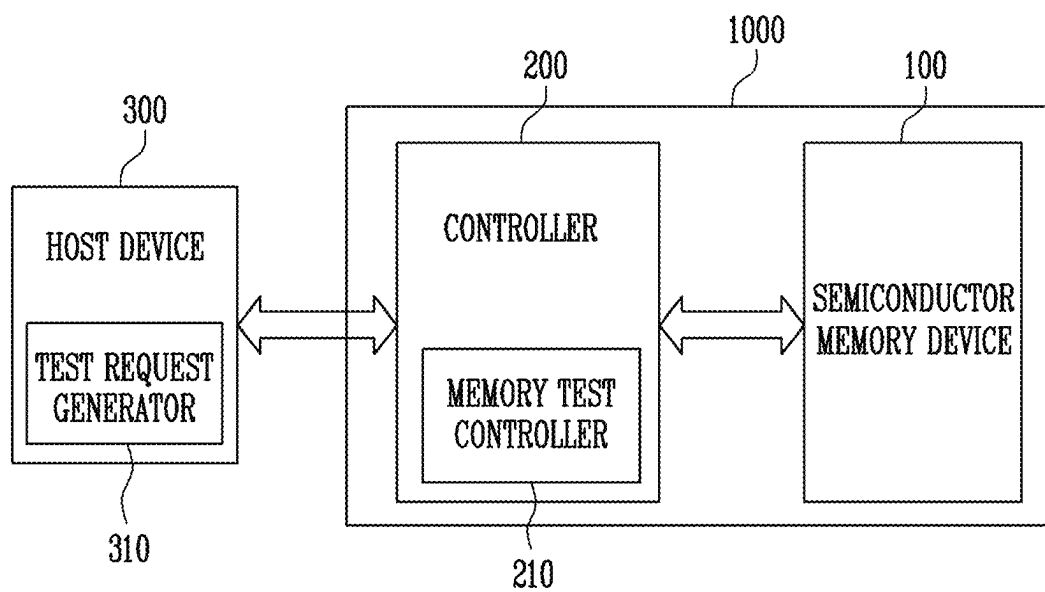


FIG. 2

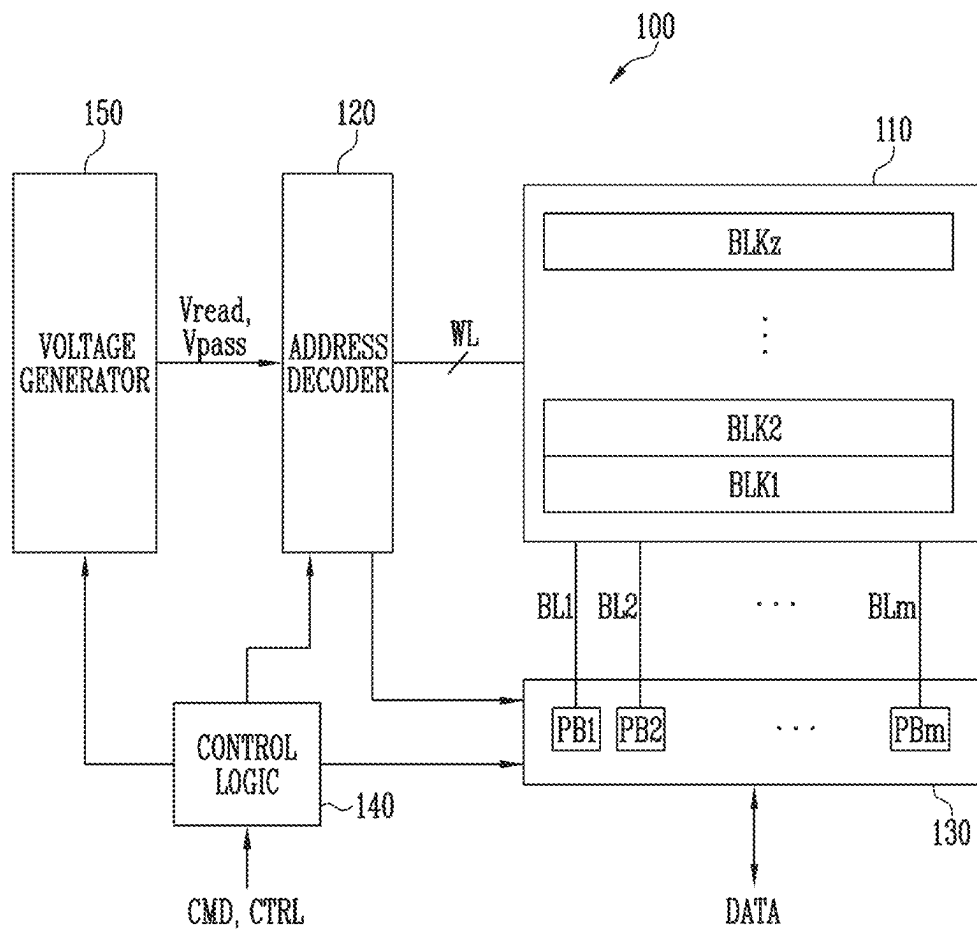


FIG. 3

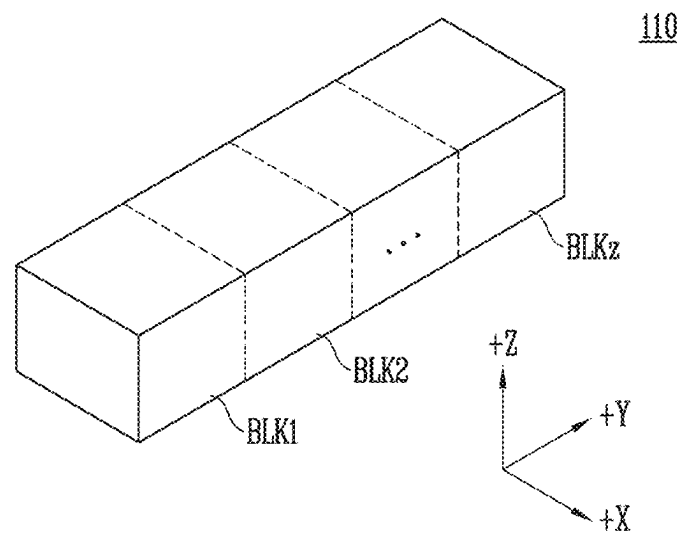


FIG. 4

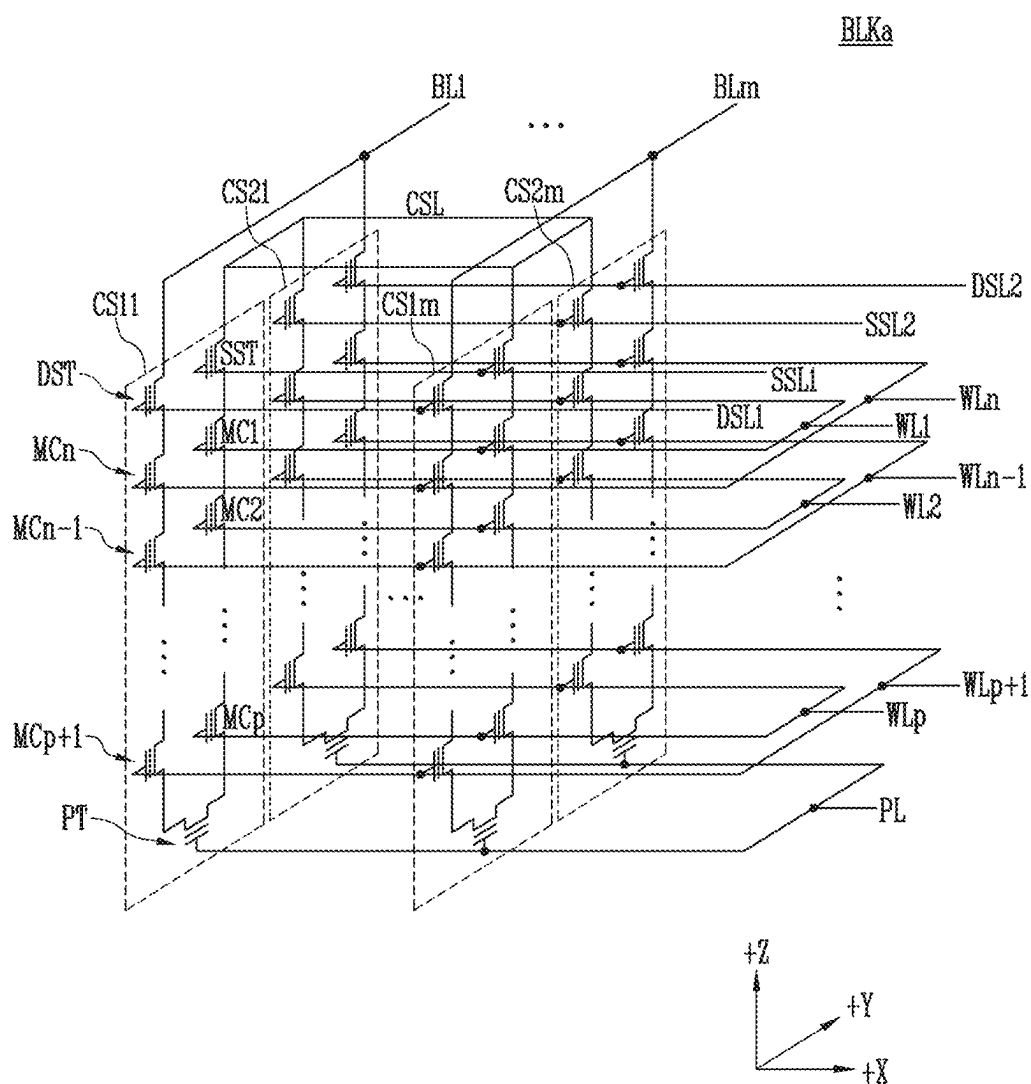


FIG. 5

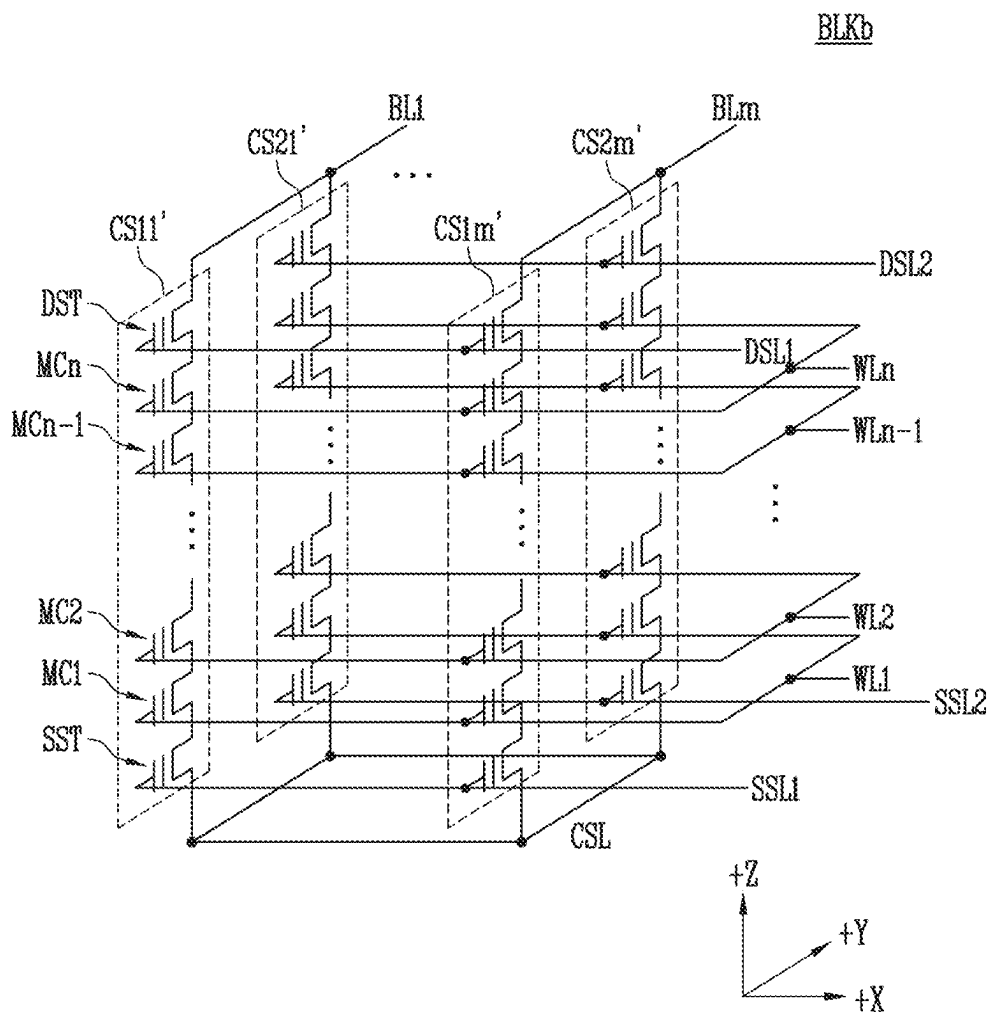


FIG. 6

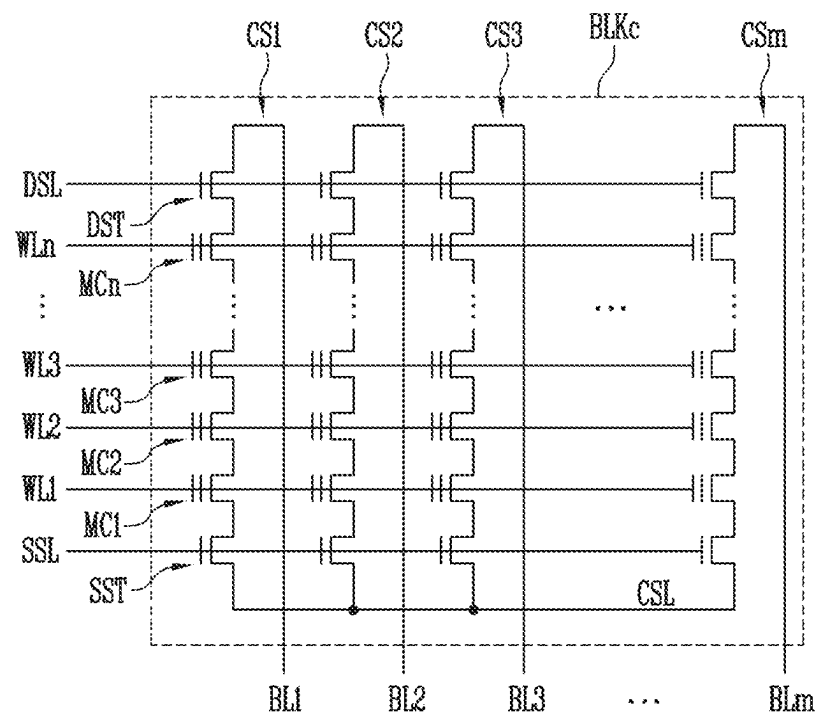


FIG. 7

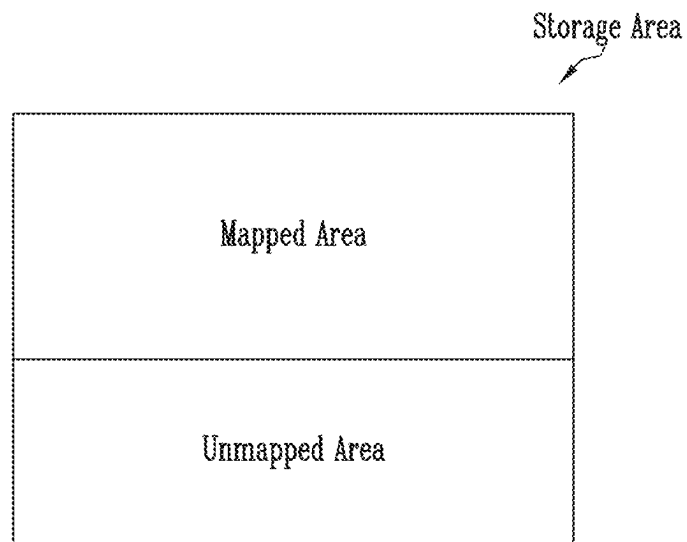


FIG. 8

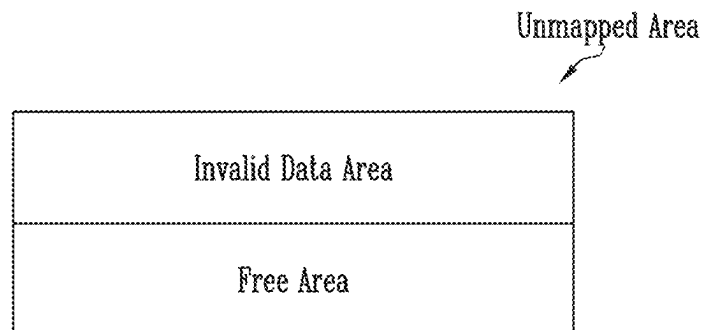


FIG. 9

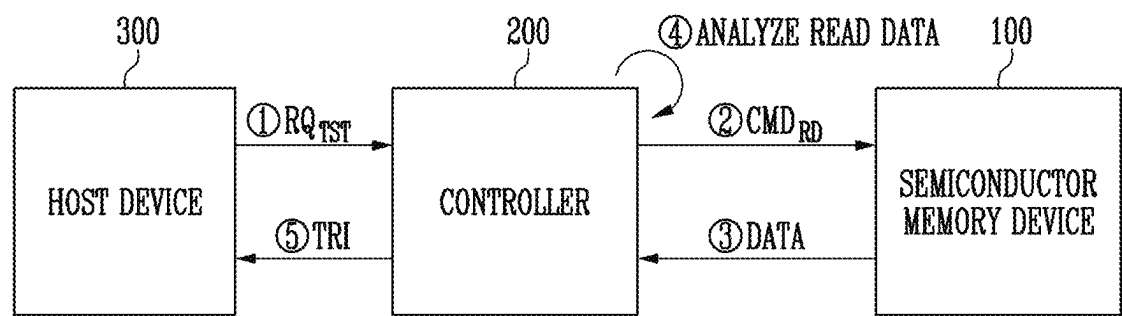


FIG. 10

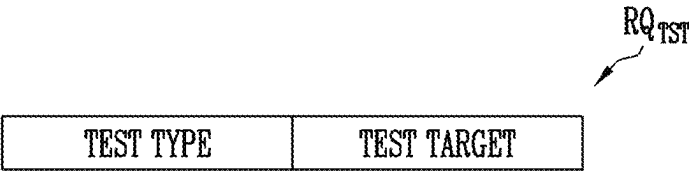


FIG. 11A

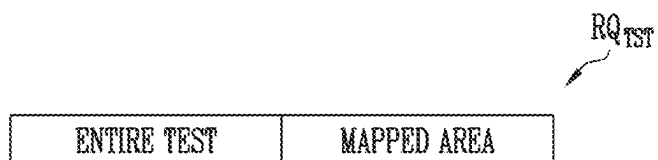


FIG. 11B

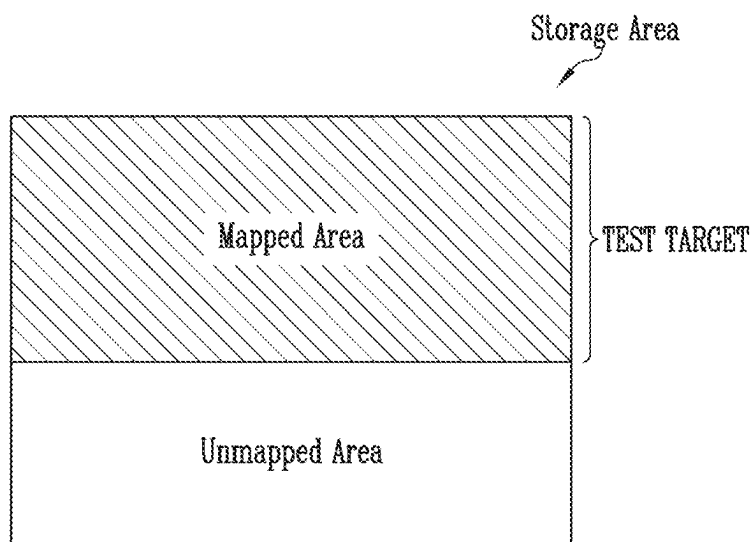


FIG. 12A

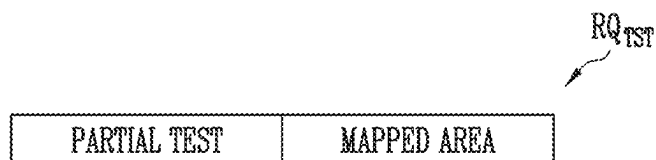


FIG. 12B

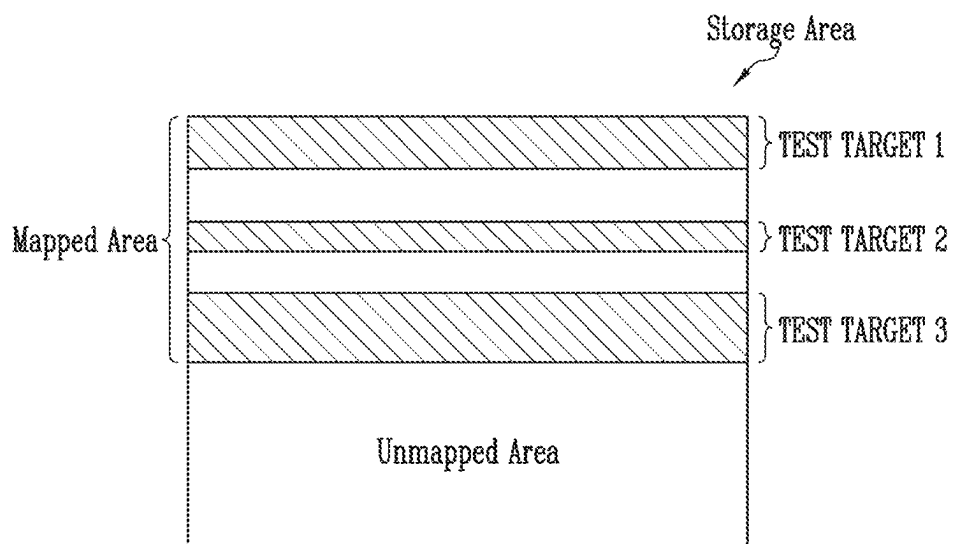


FIG. 13A

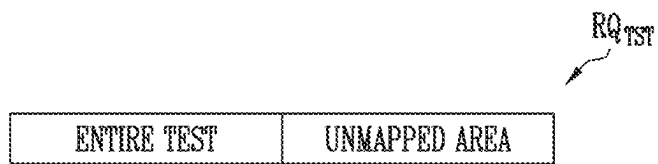


FIG. 13B

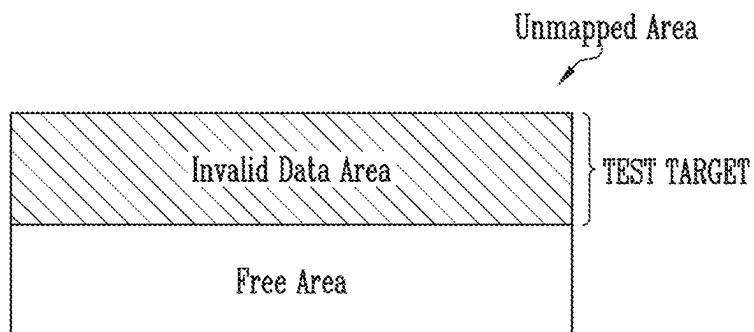


FIG. 14A

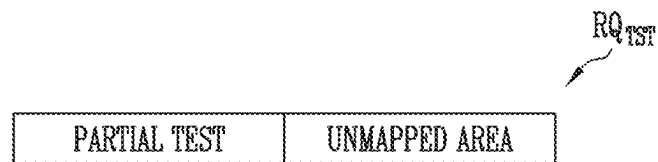


FIG. 14B

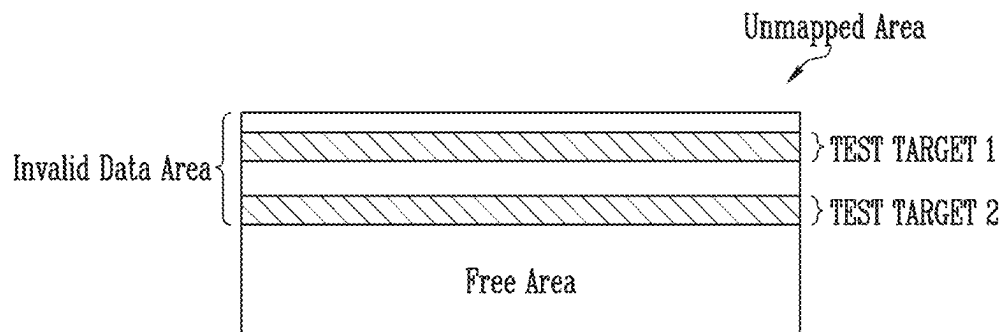


FIG. 15

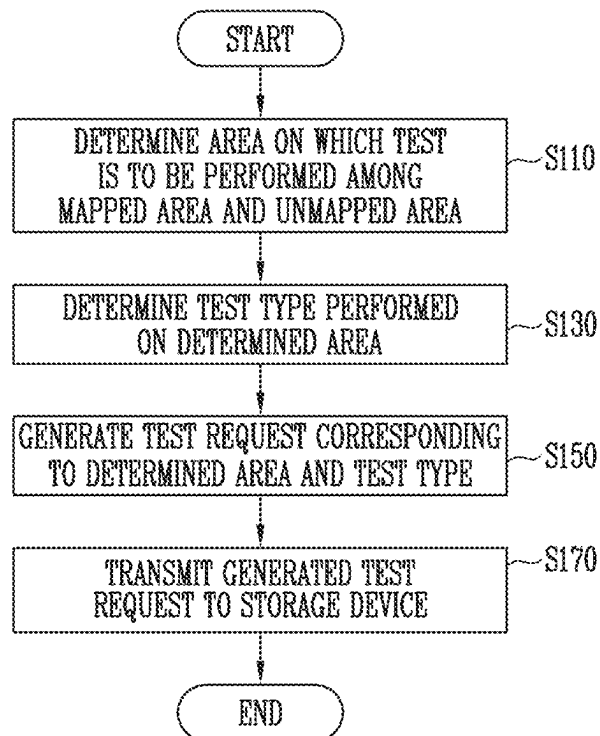


FIG. 16

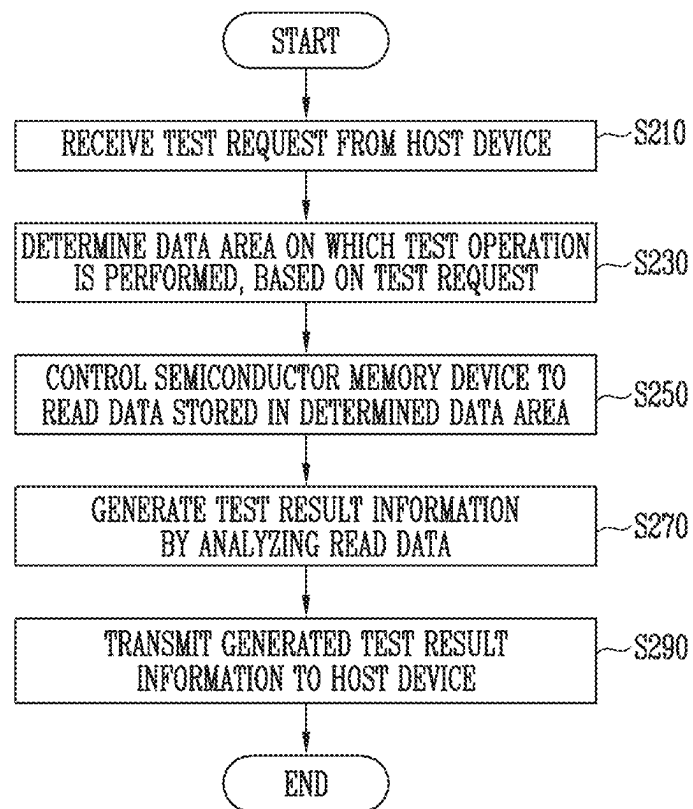


FIG. 17

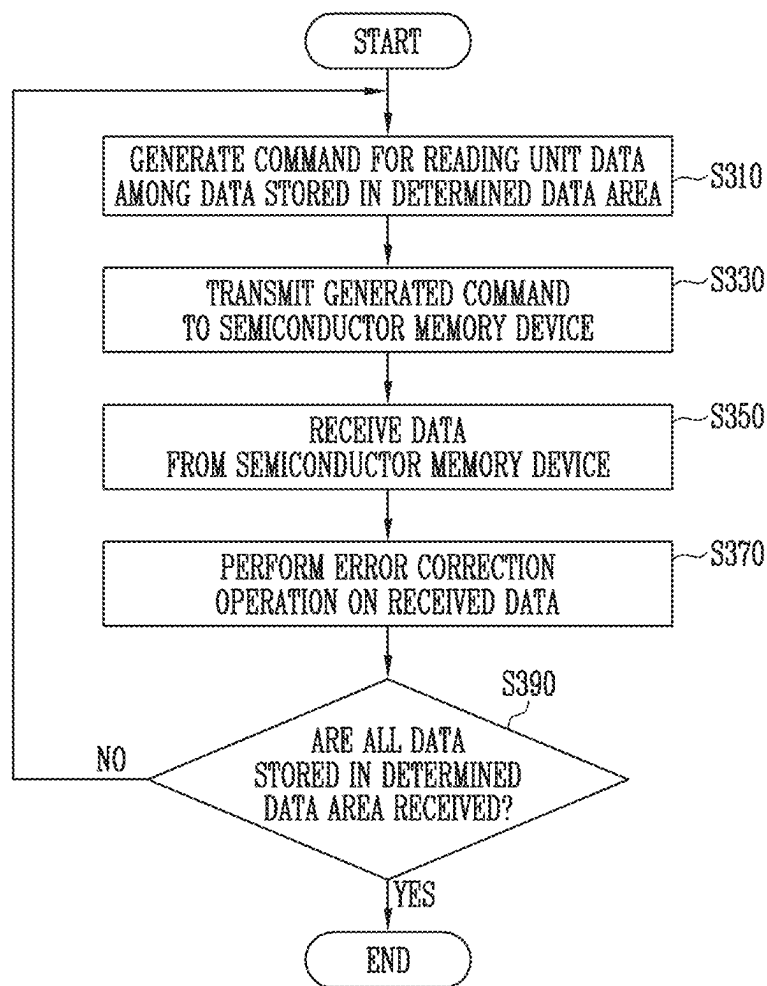


FIG. 18

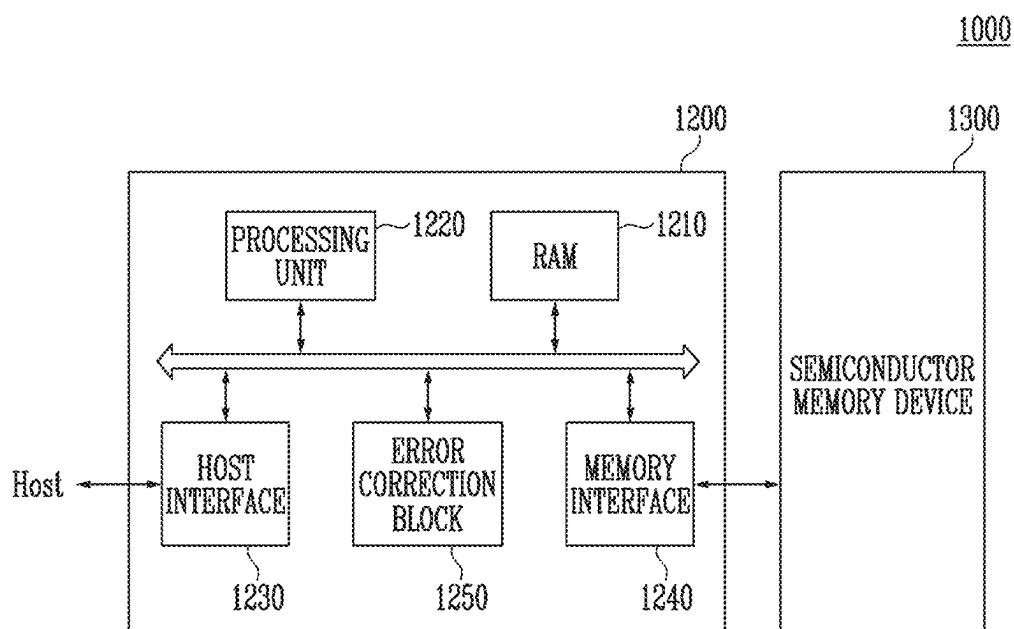


FIG. 19

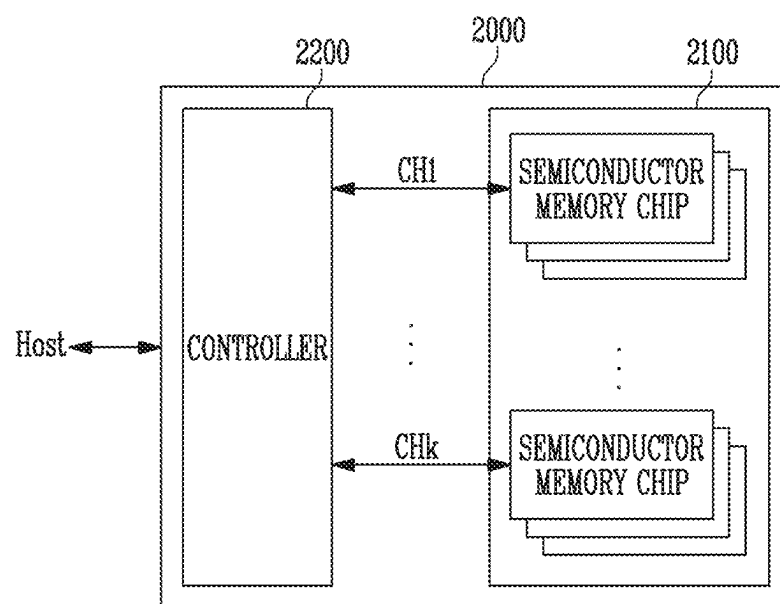
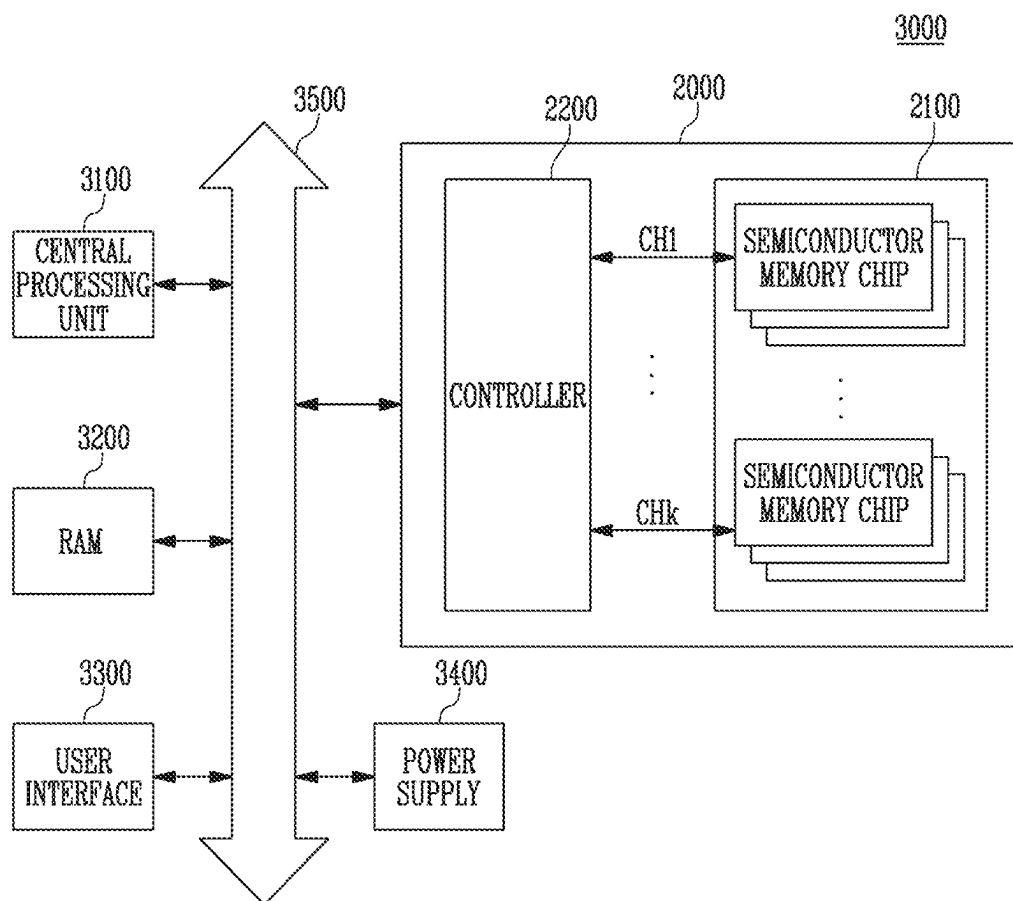


FIG. 20



STORAGE DEVICE, HOST DEVICE, AND METHOD OF OPERATING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

The present application is a continuation of U.S. patent application Ser. No. 17/748,421 filed on May 19, 2022, which is issued as U.S. Pat. No. 12,027,222 on Jul. 2, 2024, which claims priority under 35 U.S.C. § 119(a) to Korean patent application number 10-2021-0175075, filed on Dec. 8, 2021, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field of Invention

Embodiments of the present disclosure relate to an electronic device, and more particularly, to a storage device, a host device, and a method of operating the same.

2. Description of Related Art

A semiconductor memory device may be formed in a two-dimensional structure in which strings are horizontally arranged on a semiconductor substrate, or in a three-dimensional structure in which the strings are vertically stacked on the semiconductor substrate. A three-dimensional semiconductor memory device is a memory device designed to resolve a limit of an integration degree of a two-dimensional semiconductor memory device, and may include a plurality of memory cells that are vertically stacked on a semiconductor substrate. Moreover, a controller may control an operation of the semiconductor memory device according to a request from a host device.

SUMMARY

An embodiment of the present disclosure provides a storage device, a host device, and a method of operating the same capable of efficiently performing a test operation.

According to an embodiment of the present disclosure, a method of operating a host device according to the present technology includes determining an area to be tested among a mapped area and an unmapped area included in a storage area of a storage device, generating a test request corresponding to the determined area, and transmitting the generated test request to the storage device.

According to another embodiment of the present disclosure, a method of operating a storage device includes receiving a test request from an external device, determining a data area on which a test operation is performed among a mapped area and an unmapped area included in a storage area of a storage device based on the test request, controlling a semiconductor memory device included in the storage device to read data stored in the determined data area, generating test result information based on the data read by the semiconductor memory device, and transmitting the test result information to the external device.

According to still another embodiment of the present disclosure, a storage device includes a semiconductor memory device and a controller. The semiconductor memory device includes a storage area for storing data. The controller controls an operation of the semiconductor memory device in response to a request from an external device. The controller receives a test request from the

external device, determines a data area on which a test operation is performed among a mapped area and an unmapped area included in the storage area based on the test request, controls the semiconductor memory device to read data stored in the determined data area, generates test result information based on the data read by the semiconductor memory device, and transmits the test result information to the external device.

According to still another embodiment of the present disclosure, an operating method of a controller includes controlling, in response to a request from a host, a memory device to perform a read operation on at least a part of target data stored in a target area therein to measure integrity of the part read out from the target area, and providing the host with the measured integrity storage device includes a semiconductor memory device and a controller. The request indicates the target area and a range of the target data among data stored in the target region.

The present technology may provide a storage device, a host device, and a method of operating the same capable of efficiently performing a test operation.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram illustrating a storage device including a controller, and a host device according to an embodiment of the present disclosure.

FIG. 2 is a block diagram illustrating a semiconductor memory device according to an embodiment of the present disclosure.

FIG. 3 is a diagram illustrating a memory cell array of FIG. 2 according to an embodiment of the present disclosure.

FIG. 4 is a circuit diagram illustrating a memory block BLKa among memory blocks BLK1 to BLKz of FIG. 3 according to an embodiment of the present disclosure.

FIG. 5 is a circuit diagram illustrating a memory block BLKb among the memory blocks BLK1 to BLKz of FIG. 3 according to an embodiment of the present disclosure.

FIG. 6 is a circuit diagram illustrating a memory block BLKc among the plurality of memory blocks BLK1 to BLKz included in the memory cell array 110 of FIG. 2 according to an embodiment of the present disclosure.

FIG. 7 is a diagram illustrating a storage area included in the storage device according to an embodiment of the present disclosure.

FIG. 8 is a diagram illustrating an unmapped area of the storage area included in the storage device according to an embodiment of the present disclosure.

FIG. 9 is a diagram illustrating a test operation of the semiconductor memory device in response to a request from the host device according to an embodiment of the present disclosure.

FIG. 10 is a diagram illustrating the test request received from the host device according to an embodiment of the present disclosure.

FIGS. 11A and 11B are diagrams illustrating an operation of the storage device that performs an entire test on a mapped area based on the test request from the host device according to an embodiment of the present disclosure.

FIGS. 12A and 12B are diagrams illustrating an operation of the storage device that performs a partial test on the mapped area based on the test request from the host device according to an embodiment of the present disclosure.

FIGS. 13A and 13B are diagrams illustrating an operation of the storage device that performs the entire test on the

3

unmapped area based on the test request from the host device according to an embodiment of the present disclosure.

FIGS. 14A and 14B are diagrams illustrating an operation of the storage device that performs the partial test on the unmapped area based on the test request from the host device according to an embodiment of the present disclosure.

FIG. 15 is a flowchart illustrating a method of operating a host device according to an embodiment of the present disclosure.

FIG. 16 is a flowchart illustrating a method of operating a storage device according to an embodiment of the present disclosure.

FIG. 17 is a flowchart illustrating operation S250 of FIG. 16 according to an embodiment of the present disclosure.

FIG. 18 is a block diagram illustrating a storage device including a semiconductor memory device and a controller according to an embodiment of the present disclosure.

FIG. 19 is a block diagram illustrating an application example of the storage device of FIG. 18 according to an embodiment of the present disclosure.

FIG. 20 is a block diagram illustrating a computing system including the storage device described with reference to FIG. 19 according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

The advantages and features of the present disclosure, and a method of accomplishing the advantages and features will be described through embodiments that are described in detail below together with the accompanying drawings. However, the present disclosure is not limited to the embodiments described herein but may be embodied in other forms. The present embodiments are provided to describe the technical spirit of the present disclosure in detail to those skilled in the art to which the present disclosure pertains so that those skilled in the art may easily implement the technical spirit of the present disclosure.

FIG. 1 is a block diagram illustrating a storage device 1000 including a controller 200, and a host device 300 according to an embodiment of the present disclosure.

Referring to FIG. 1, the storage device 1000 includes a semiconductor memory device 100 and a controller 200. In addition, the storage device 1000 communicates with an external device. In an embodiment, the external device may be the host device 300. In the present disclosure, the storage device 1000 communicates with the host device 300, but the present disclosure is not limited thereto. That is, in addition to the host device 300, other types of devices communicating with the storage device 1000 may also be included in the external device.

The controller 200 controls an overall operation of the semiconductor memory device 100. In addition, the controller 200 controls an operation of the semiconductor memory device 100 based on an operation request received from the host device 300.

The semiconductor memory device 100 operates under control of the controller 200. The semiconductor memory device 100 includes a memory cell array 110 having a plurality of memory blocks. In an embodiment, the semiconductor memory device 100 may be a non-volatile memory device. For example, the semiconductor memory device 100 may include at least one of a flash memory device, a phase-change random access memory (PCRAM),

4

a magnetoresistive random access memory (MRAM), a resistive random access memory (ReRAM), and the like.

The controller 200 may receive a write request, a read request, a trim request, or the like from the host device 300, and control the semiconductor memory device 100 based on the received requests. More specifically, the controller 200 may generate commands for controlling the operation of the semiconductor memory device 100 and transmit the commands to the semiconductor memory device 100.

The semiconductor memory device 100 is configured to receive a command and an address from the controller 200 and to access an area selected by the address of the memory cell array. That is, the semiconductor memory device 100 performs an internal operation corresponding to a command on the area selected by the address.

For example, the semiconductor memory device 100 may perform a program operation, a read operation, and an erase operation. During the program operation, the semiconductor memory device 100 may program data in the area selected by the address. During the read operation, the semiconductor memory device 100 may read data from the area selected by the address. During the erase operation, the semiconductor memory device 100 may erase data stored in the area selected by the address.

The host device 300 includes a test request generator 310. The test request generator 310 may generate a test request for controlling the storage device 1000 to perform an internal test operation. The generated test request may be transmitted to the controller 200 of the storage device 1000. The controller 200 may control the semiconductor memory device 100 to perform a test operation on the memory cell array 110 of the semiconductor memory device 100 based on the received test request.

As an example, the host device 300 may generate a self-test request. The self-test request is a request for controlling the storage device 1000 to independently perform the test operation without test equipment. The storage device 1000 may perform a self-test operation in response to the self-test request received from the host device 300. The self-test request for the storage device 1000 may be used for the following purposes.

- Start a short self-test operation (short self-test).
- Start an extended self-test operation (extended self-test).
- Start a vendor specific self-test operation (vendor specific self-test).
- Abort a self-test operation that is already in progress.

The self-test operation may be performed by the controller 200 of the storage device 1000. The self-test operation may be a diagnostic test sequence for testing integrity and functionality of the controller 200 and may include a media test related to a namespace. The self-test operation may be divided into a series of segments, and each segment may be configured of a set of vendor specific tests. A segment number of a "self-test result data structure" may be used for reporting purposes to display a failure when a test is failed. In the short self-test operation and the extended self-test operation, the tests performed in each segment may be the same.

According to that described in a basic specification of NVMe, as specified in Table 1 below, a namespace identifier field may control the namespace included in the self-test operation of the storage device 1000. In order to distinguish a command from the command transmitted from the controller 200 to the semiconductor memory device 100, a term "command" described in the NVMe is replaced with "request" in the present specification.

5

TABLE 1

Value	Description
00000000h	Specifies that the self-test operation does not include the namespace and only the controller is included as a portion of the self-test operation.
00000001h~FFFFFFEh	Specifies that the self-test operation includes the namespace specified in this field. When this field specifies an invalid namespace ID, the controller is required to abort the request with a status of an invalid namespace or format. When this field specifies an inactive namespace ID, the controller is required to abort the request with a status of an invalid field in the request.
FFFFFFFFh	Specifies that when the self-test operation is started, a device independent self-test operation includes all active namespaces accessible through the controller.

As an example, as disclosed in Table 2 below, the self-test request may use a command Dword 10 field. In this case, all other command specific fields may be reserved.

TABLE 2

Bits	Description
31:04	Reserved.
03:00	Self-test code (STC): This field specifies an operation performed by the self-test request.

More specifically, the STC specified by the bits (03:00) of Table 2 is shown in Table 3 below.

TABLE 3

Value	Definition
0h	Reserved
1h	Start the short self-test operation (short self-test)
2h	Start the extended self-test operation (extended self-test)
3h~Dh	Reserved
Eh	Vendor specific
Fh	Abort the self-test operation

In an embodiment, according to that described in the basic specification of the NVMe, a method of processing a self-test command received according to an operation status of the storage device **1000** is defined in Table 4 below.

TABLE 4

Self-test progress status	STC	Controller operation
Yes	1h	Abort a new self-test request as a state in which the self-test operation is in progress.
	2h	Vendor Specific.
	Eh	The controller sequentially performs the following operations.
	Fh	1. Stop the self-test operation that is in progress 2. Generate a log entry in the “newest self-test result data structure” in a self-test log. 3. Set a “current device self-test status” field of a “device self-test log” to 0h 4. Successfully complete the request
No	1h	The controller sequentially performs the following operations.
		1. Check a parameter of the request 2. Set the “current device self-test status” field of the “device self-test log” to 1h

6

TABLE 4-continued

Self-test progress status	STC	Controller operation
5	2h	3. Start the self-test operation 4. Successfully complete the request The controller sequentially performs the following operations.
		1. Check the parameter of the request 2. Set the “current device self-test status” field of the “device self-test log” to 2h 3. Start the self-test operation 4. Successfully complete the request
10	Eh Fh	Vendor specific. Successfully complete the request. The “device self-test log” is not corrected.
15		

The storage device **1000** may maintain a log page related to the self-test. For example, the storage device **1000** may have a log identifier of a log page “06 h” related to the self-test, and may indicate the following.

- The status of the device independent self-test task that is in progress and a completion rate of the corresponding task.
- The last 20 self-test operation results.

The self-test result data structure included in the “newest self-test result data structure” field is always the result of the last completed or aborted self-test operation. The next self-test result data structure field includes the result of the second most recent self-test operation. For example, when less than 20 self-test tasks are completed or aborted, a self-test status field of an unused self-test result data structure field may be required to be set to Fh, and all other fields in the corresponding self-test result data structure are ignored. As an example, the self-test log may be configured as shown in Table 5 below.

TABLE 5

Bytes	Description
40	00
	Current self-test operation: This field defines the current self-test operation of the storage device. Bit 7:4 is reserved. Bit 3:0 indicates the status of the current self-test operation as defined in Table 6 below. When the self-test operation is in progress (that is, this field is set to 1h or 2h), the controller is required not to set this field to 0h until the new self-test result data structure is generated (that is, when the self-test operation of the storage device 1000 is completed or aborted, the controller is required to generate the self-test result data structure before setting this field to 0h).
50	01
	Current self-test completion: This field defines the completion status of the current self-test. Bit 7 is reserved. Bit 6:0 indicates a percentage of the completed self-test operation (for example, a value of 25 indicates that 25% of the self-test operation is completed and 75% remains). When bit 3:0 of the current self-test operation field is vacated to 0h (indicating that the self-test operation that is in progress does not exist), this field is ignored.
55	03:02
	Reserved
60	31:04
	The newest self-test result data structure (refer to Table 7 below)
65	59:32
	Second most recent self-test result data structure (refer to Table 7 below)
65	535:508
	Nineteenth most recent self-test result data structure (refer to Table 7 below)
65	563:536
	Twentieth most recent self-test result data structure (refer to Table 7 below)

7

TABLE 6

Value	Definition
0h	The self-test operation that is in progress does not exist.
1h	The short self-test operation is in progress.
2h	The extended self-test operation is in progress.
3h~Dh	Reserved
Eh	Vendor Specific
Fh	Reserved

TABLE 7

Bytes	Description
00	Self-test status: This field indicates the self-test code and the operation status. Bit 7:4 indicates a self-test code value specified in the self-test request starting the self-test operation described by this self-test result data structure (refer to Table 3). Bit 3:0 indicates the result of the device independent self-test task described by this self-test result data structure (refer to Table 8).
01	Segment number: This field indicates a segment number where a first self-test failure occurs. This field is ignored when the self-test status field bit [3:0] is not set to 7h.
02	Valid diagnostic information: This field indicates reported diagnostic error information. Bit 3 (SC valid): When the bit 3 is set to '1', the bit 3 indicates that a content of the status code (SC) field is valid. When the bit 3 is vacated to '0', the bit 3 indicates that the content of the SC field is invalid. Bit 2 (SCT valid): When the bit 2 is set to '1', the bit 2 indicates that a content of the status code type (SCT) field is valid. When the bit 2 is vacated to '0', the bit 2 indicates that the content of the SCT field is invalid. Bit 1 (FLBA valid): When the bit 1 is set to '1', the bit 1 indicates that a content of a failed LBA (FLBA) field is valid. When the bit 1 is vacated to '0', the bit 1 indicates that the content of the FLBA field is invalid. Bit 0 (NSID Valid): When the bit 0 is set to '1', the bit 0 indicates that a content of a namespace identifier (NSID) field is valid. When the bit 0 is vacated to '0', the bit 0 indicates that the content of the NSID field is invalid.
03	Reserved
11:04	Power on hours (POH): This field indicates a power-on time of the storage device 1000 when the self-test operation is completed or aborted. The POH does not include a time when power is supplied to the controller and the controller is in a low power status.
15:12	Namespace identifier (NSID): This field indicates the namespace in which a failed LBA occurs. A content of this field is only valid when an NSID valid bit is set to '1'.
23:16	Failing LBA (FLBA): This field indicates an LBA of a logical block that causes the test to be failed. When the device finds two or more failed logical blocks during the test, this field indicates only one of the failed logical blocks. A content of this field is only valid when the FLBA valid bit is set to '1'.
24	Status code type (SCT): This field may include additional information related to an error or a condition. Bit 7:3 is reserved. Bit 2:0 may include additional information related to the error or the condition generated during the self-test operation, displayed in the same format used in an SCT field of a completion queue entry. A content of this field is valid only when the SCT valid bit is set to '1'.
25	Status Code: This field may include additional information related to the error or the condition generated during the self-test operation, displayed in the same format used in the status code (SC) field of

8

TABLE 7-continued

Bytes	Description
5	the completion queue entry. A content of this field is valid only when the SC valid bit is set to '1'.
27:26	Vendor specific

TABLE 8

Value	Definition
0h	The task is completed without an error.
1h	The operation is aborted due to the self-test request.
2h	The operation is aborted due to a controller level reset.
3h	The operation is aborted because the namespace is removed from a namespace inventory.
4h	The operation is aborted due to processing of an NVM format request.
5h	A fatal or unknown test error occurs while the controller performs the self-test operation, and the operation is not completed.
6h	The operation is completed, but a failed segment occurs, and the failed segment is unknown.
7h	The operation is completed, but one or more failed segments occur, and the failed first segment is displayed in the segment number field.
8h	The task is aborted due to an unknown reason.
9h	The task is aborted due to a sanitize task.
Ah~Eh	Reserved
Fh	The entry is not used (the test result is not included)

The self-test operation may be performed in a background. Accordingly, the self-test operation may be processed simultaneously with some requests, and the self-test operation may be required to be temporarily aborted to process another request. When receiving a request to temporarily abort the self-test operation during the self-test operation, the controller **200** aborts the self-test operation, processes and completes an operation related to the received request, and then resumes the self-test operation. During the self-test operation, performance of a subsystem of the storage device **1000** may be degraded.

As an example, the short self-test operation is required to be completed within 2 minutes. A completion rate of the short self-test operation may be displayed in a current completion rate field of a self-test log (refer to Table 5).

An abortion condition of the simple self-test operation is as follows.

- The self-test operation is required to be aborted by a controller level reset affecting the performed controller.
- The self-test operation is required to be aborted by an NVM format command.
- The self-test operation is required to be aborted when a sanitize operation is started.
- The self-test operation is required to be aborted when a self-test request in which the self-test code field is set to Fh is processed.
- The self-test operation may be aborted when a designated namespace is removed from the namespace inventory.

The extended self-test operation is required to be completed within a time displayed in an "extended self-test time" field of a "controller identify controller" data structure. The completion rate of the extended self-test operation may be displayed in a current completion rate field of the self-test log (refer to Table 5).

The extended self-test operation is required to persist also in a case where all controller level resets exist and is required to be restarted after the reset is completed or power recovery exists.

An abortion condition of the extensive self-test operation is as follows.

- a) The extended self-test operation is required to be aborted by the NVM format command.
- b) The extended self-test operation is required to be aborted when the sanitize operation is started.
- c) The extended self-test operation is required to be aborted when a self-test request in which the self-test code field is set to Fh is processed.
- d) The extended self-test operation may be aborted when the designated namespace is removed from the namespace inventory.

More specifically, the controller **200** includes a memory test controller **210**. The memory test controller **210** receives the test request generated by the test request generator **310** of the host **300**. The memory test controller **210** controls the test operation of the semiconductor memory device **100** based on the received test request. More specifically, the memory test controller **210** may generate at least one command for controlling the semiconductor memory device **100** to perform the test operation. The command generated by the memory test controller **210** is transmitted to the semiconductor memory device **100**. The semiconductor memory device **100** may perform the test operation in response to the received command.

FIG. 2 is a block diagram illustrating a semiconductor memory device according to an embodiment of the present disclosure.

Referring to FIG. 2, the semiconductor memory device **100** includes a memory cell array **110**, an address decoder **120**, a read and write circuit **130**, a control logic **140**, and a voltage generator **150**.

The memory cell array **110** includes a plurality of memory blocks BLK1 to BLKz. The plurality of memory blocks BLK1 to BLKz are connected to the address decoder **120** through word lines WL. The plurality of memory blocks BLK1 to BLKz are connected to the read and write circuit **130** through bit lines BL1 to BLm. Each of the plurality of memory blocks BLK1 to BLKz includes a plurality of memory cells. In an embodiment, the plurality of memory cells are non-volatile memory cells, and may be configured of non-volatile memory cells having a vertical channel structure. The memory cell array **110** may be configured as a memory cell array of a two-dimensional structure. According to an embodiment, the memory cell array **110** may be configured as a memory cell array of a three-dimensional structure. Each of the plurality of memory cells included in the memory cell array may store at least one bit of data. In an embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a single-level cell (SLC) storing one bit of data. In another embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a multi-level cell (MLC) storing two bits of data. In still another embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a triple-level cell storing three bits of data. In still another embodiment, each of the plurality of memory cells included in the memory cell array **110** may be a quad-level cell storing four bits of data. According to an embodiment, the memory cell array **110** may include a plurality of memory cells each storing five or more bits of data.

The address decoder **120**, the read and write circuit **130**, the control logic **140**, and the voltage generator **150** operate as a peripheral circuit that drives the memory cell array **110**. The address decoder **120** is connected to the memory cell array **110** through the word lines WL. The address decoder

120 is configured to operate in response to control of the control logic **140**. The address decoder **120** receives an address through an input/output buffer (not shown) inside the semiconductor memory device **100**.

The address decoder **120** is configured to decode a block address among received addresses. The address decoder **120** selects at least one memory block according to the decoded block address. In addition, the address decoder **120** applies a read voltage Vread generated by the voltage generator **150** to a selected word line among the selected memory block at a read voltage application operation during a read operation, and applies a pass voltage Vpass to the remaining unselected word lines. In addition, the address decoder **120** applies a verify voltage generated by the voltage generator **150** to the selected word line among the selected memory block and applies the pass voltage Vpass to the remaining unselected word lines during a program verify operation.

The address decoder **120** is configured to decode a column address of the received addresses. The address decoder **120** transmits the decoded column address to the read and write circuit **130**.

The read operation and a program operation of the semiconductor memory device **100** are performed in a page unit. Addresses received at a time of a request of the read operation and the program operation include a block address, a row address, and a column address. The address decoder **120** selects one memory block and one word line according to the block address and the row address. The column address is decoded by the address decoder **120** and is provided to the read and write circuit **130**.

The address decoder **120** may include a block decoder, a row decoder, a column decoder, an address buffer, and the like.

The read and write circuit **130** includes a plurality of page buffers PB1 to PBm. The read and write circuit **130** may operate as a "read circuit" during a read operation of the memory cell array **110** and may operate as a "write circuit" during a write operation of the memory cell array **110**. The plurality of page buffers PB1 to PBm are connected to the memory cell array **110** through the bit lines BL1 to BLm. During the read operation and the program verify operation, in order to sense a threshold voltage of the memory cells, the plurality of page buffers PB1 to PBm sense a change of an amount of a current flowing according to a program state of a corresponding memory cell through a sensing node while continuously supplying a sensing current to the bit lines connected to the memory cells, and latches the sensed change as sensing data. The read and write circuit **130** operates in response to page buffer control signals output from the control logic **140**.

During the read operation, the read and write circuit **130** senses data of the memory cell, temporarily stores read data, and outputs data DATA to the input/output buffer (not shown) of the semiconductor memory device **100**. In an embodiment, the read and write circuit **130** may include a column selection circuit, and the like, in addition to the page buffers (or page registers).

The control logic **140** is connected to the address decoder **120**, the read and write circuit **130**, and the voltage generator **150**. The control logic **140** receives a command CMD and a control signal CTRL through the input/output buffer (not shown) of the semiconductor memory device **100**. The control logic **140** is configured to control overall operations of the semiconductor memory device **100** in response to the control signal CTRL. In addition, the control logic **140** outputs a control signal for adjusting a sensing node pre-charge potential level of the plurality of page buffers PB1 to

PBm. The control logic 140 may control the read and write circuit 130 to perform the read operation of the memory cell array 110.

The voltage generator 150 generates the read voltage V_{read} and the pass voltage V_{pass} during the read operation in response to the control signal output from the control logic 140. In order to generate a plurality of voltages having various voltage levels, the voltage generator 150 may include a plurality of pumping capacitors that receive an internal power voltage, and generate the plurality of voltages by selectively activating the plurality of pumping capacitors in response to the control of the control logic 140.

The address decoder 120, the read and write circuit 130, and the voltage generator 150 may function as a "peripheral circuit" that performs the read operation, the write operation, and an erase operation on the memory cell array 110. The peripheral circuit performs the read operation, the write operation, and the erase operation on the memory cell array 110 based on the control of the control logic 140.

FIG. 3 is a diagram illustrating the memory cell array of FIG. 2 according to an embodiment of the present disclosure.

Referring to FIG. 3, the memory cell array 110 includes a plurality of memory blocks BLK1 to BLKz. Each memory block may have a three-dimensional structure. Each memory block includes a plurality of memory cells stacked on a substrate. The plurality of memory cells are arranged along a +X direction, a +Y direction, and a +Z direction. A structure of each memory block is described in more detail with reference to FIGS. 4 and 5.

FIG. 4 is a circuit diagram illustrating a memory block BLKa among the memory blocks BLK1 to BLKz of FIG. 3 according to an embodiment of the present disclosure.

Referring to FIG. 4, the memory block BLKa includes a plurality of cell strings CS11 to CS1m and CS21 to CS2m. In an embodiment, each of the plurality of cell strings CS11 to CS1m and CS21 to CS2m may be formed in a 'U' shape. In the memory block BLKa, m cell strings are arranged in a row direction, that is, the +X direction. In FIG. 4, two cell strings are arranged in a column direction, that is, the +Y direction. However, this is for convenience of description and it may be understood that three or more cell strings may be arranged in the column direction.

Each of the plurality of cell strings CS11 to CS1m and CS21 to CS2m includes at least one source select transistor SST, first to n-th memory cells MC1 to MCn, a pipe transistor PT, and at least one drain select transistor DST.

Each of the select transistors SST and DST and the memory cells MC1 to MCn may have a similar structure. In an embodiment, each of the select transistors SST and DST and the memory cells MC1 to MCn may include a channel layer, a tunneling insulating layer, a charge storage layer, and a blocking insulating layer. In an embodiment, a pillar for providing the channel layer may be provided in each cell string. In an embodiment, a pillar for providing at least one of the channel layer, the tunneling insulating layer, the charge storage layer, and the blocking insulating layer may be provided in each cell string.

The source select transistor SST of each cell string is connected between a common source line CSL and the memory cells MC1 to MCp.

In an embodiment, the source select transistors of the cell strings arranged in the same row are connected to a source select line extending in the row direction, and the source select transistors of the cell strings arranged in different rows are connected to different source select lines. In FIG. 4, the source select transistors of the cell strings CS11 to CS1m of

a first row are connected to a first source select line SSL1. The source select transistors of the cell strings CS21 to CS2m of a second row are connected to a second source select line SSL2.

In another embodiment, the source select transistors of the cell strings CS11 to CS1m and CS21 to CS2m may be commonly connected to one source select line.

The first to n-th memory cells MC1 to MCn of each cell string are connected between the source select transistor SST and the drain select transistor DST.

The first to n-th memory cells MC1 to MCn may be divided into first to p-th memory cells MC1 to MCp and (p+1)-th to n-th memory cells MCp+1 to MCn. The first to p-th memory cells MC1 to MCp are sequentially arranged in a direction opposite to the +Z direction, and are connected in series between the source select transistor SST and the pipe transistor PT. The (p+1)-th to n-th memory cells MCp+1 to MCn are sequentially arranged in the +Z direction, and are connected in series between the pipe transistor PT and the drain select transistor DST. The first to p-th memory cells MC1 to MCp and the (p+1)-th to n-th memory cells MCp+1 to MCn are connected to each other through the pipe transistor PT. Gates of the first to n-th memory cells MC1 to MCn of each cell string are connected to the first to n-th word lines WL1 to WLn, respectively.

A gate of the pipe transistor PT of each cell string is connected to a pipeline PL.

The drain select transistor DST of each cell string is connected between a corresponding bit line and the memory cells MCp+1 to MCn. Cell strings arranged in the row direction are connected to the drain select line extending in the row direction. The drain select transistors of the cell strings CS11 to CS1m of the first row are connected to a first drain select line DSL1. The drain select transistors of the cell strings CS21 to CS2m of the second row are connected to a second drain select line DSL2.

The cell strings arranged in the column direction are connected to the bit lines extending in the column direction. In FIG. 4, the cell strings CS11 and CS21 of the first column are connected to the first bit line BL1. The cell strings CS1m and CS2m of the m-th column are connected to the m-th bit line BLm.

The memory cells connected to the same word line in the cell strings arranged in the row direction configure one page. For example, the memory cells connected to the first word line WL1, among the cell strings CS11 to CS1m of the first row configure one page. The memory cells connected to the first word line WL1, among the cell strings CS21 to CS2m of the second row configure another page. The cell strings arranged in one row direction may be selected by selecting one of the drain select lines DSL1 and DSL2. One page of the selected cell strings may be selected by selecting one of the word lines WL1 to WLn.

In another embodiment, even bit lines and odd bit lines may be provided instead of the first to m-th bit lines BL1 to BLm. In addition, even-numbered cell strings among the cell strings CS11 to CS1m or CS21 to CS2m arranged in the row direction may be connected to the even bit lines, and odd-numbered cell strings among the cell strings CS11 to CS1m or CS21 to CS2m arranged in the row direction may be connected to odd bit lines, respectively.

In an embodiment, at least one of the first to n-th memory cells MC1 to MCn may be used as a dummy memory cell. For example, at least one dummy memory cell is provided to reduce an electric field between the source select transistor SST and the memory cells MC1 to MCp. Alternatively, at least one dummy memory cell is provided to reduce an

13

electric field between the drain select transistor DST and the memory cells MC_{p+1} to MC_n. As more dummy memory cells are provided, reliability of an operation for the memory block BLK_a is improved, however, the size of the memory block BLK_a increases. As less memory cells are provided, the size of the memory block BLK_a may be reduced, however, the reliability of the operation for the memory block BLK_a may be reduced.

In order to efficiently control at least one dummy memory cell, each of the dummy memory cells may have a required threshold voltage. Before or after an erase operation for the memory block BLK_a, program operations for all or a part of the dummy memory cells may be performed. When the erase operation is performed after the program operation is performed, the dummy memory cells may have the required threshold voltage by controlling a voltage applied to dummy word lines connected to the respective dummy memory cells.

FIG. 5 is a circuit diagram illustrating a memory block BLK_b among the memory blocks BLK₁ to BLK_z of FIG. 3 according to an embodiment of the present disclosure.

Referring to FIG. 5, the memory block BLK_b includes a plurality of cell strings CS_{11'} to CS_{1m'} and CS_{21'} to CS_{2m'}. Each of the plurality of cell strings CS_{11'} to CS_{1m'} and CS_{21'} to CS_{2m'} extends along a +Z direction. Each of the plurality of cell strings CS_{11'} to CS_{1m'} and CS_{21'} to CS_{2m'} includes at least one source select transistor SST, first to n-th memory cells MC₁ to MC_n, and at least one drain select transistor DST stacked on a substrate (not shown) under the memory block BLK_b.

The source select transistor SST of each cell string is connected between a common source line CSL and memory cells MC₁ to MC_n. The source select transistors of the cell strings arranged in the same row are connected to the same source select line. The source select transistors of the cell strings CS_{11'} to CS_{1m'} arranged in a first row are connected to a first source select line SSL₁. The source select transistors of the cell strings CS_{21'} to CS_{2m'} arranged in a second row are connected to a second source select line SSL₂. In another embodiment, the source select transistors of the cell strings CS_{11'} to CS_{1m'} and CS_{21'} to CS_{2m'} may be commonly connected to one source select line.

The first to n-th memory cells MC₁ to MC_n of each cell string are connected in series between the source select transistor SST and the drain select transistor DST. Gates of the first to n-th memory cells MC₁ to MC_n are connected to first to the n-th word lines WL₁ to WL_n, respectively.

The drain select transistor DST of each cell string is connected between a corresponding bit line and the memory cells MC₁ to MC_n. The drain select transistors of the cell strings arranged in the row direction are connected to a drain select line extending in the row direction. The drain select transistors of the cell strings CS_{11'} to CS_{1m'} of a first row are connected to a first drain select line DSL₁. The drain select transistors of the cell strings CS_{21'} to CS_{2m'} of a second row are connected to a second drain select line DSL₂.

As a result, the memory block BLK_b of FIG. 5 has an equivalent circuit similar to that of the memory block BLK_a of FIG. 4 except that the pipe transistor PT is excluded from each cell string.

In another embodiment, even bit lines and odd bit lines may be provided instead of the first to m-th bit lines BL₁ to BL_m. In addition, even-numbered cell strings among the cell strings CS_{11'} to CS_{1m'} or CS_{21'} to CS_{2m'} arranged in the row direction may be connected to even bit lines, and odd-numbered cell strings among the cell strings CS_{11'} to

14

CS_{1m'} or CS_{21'} to CS_{2m'} arranged in the row direction may be connected to odd bit lines, respectively.

In an embodiment, at least one of the first to n-th memory cells MC₁ to MC_n may be used as a dummy memory cell. For example, at least one dummy memory cell is provided to reduce an electric field between the source select transistor SST and the memory cells MC₁ to MC_n. Alternatively, at least one dummy memory cell is provided to reduce an electric field between the drain select transistor DST and the memory cells MC₁ to MC_n. As more dummy memory cells are provided, reliability of an operation for the memory block BLK_b is improved, however, the size of the memory block BLK_b increases. As less memory cells are provided, the size of the memory block BLK_b may be reduced, however, the reliability of the operation for the memory block BLK_b may be reduced.

In order to efficiently control at least one dummy memory cell, each of the dummy memory cells may have a required threshold voltage. Before or after an erase operation for the memory block BLK_b, program operations for all or a part of the dummy memory cells may be performed. When the erase operation is performed after the program operation is performed, the dummy memory cells may have the required threshold voltage by controlling a voltage applied to the dummy word lines connected to the respective dummy memory cells.

FIG. 6 is a circuit diagram illustrating a memory block BLK_c among the plurality of memory blocks BLK₁ to BLK_z included in the memory cell array 110 of FIG. 2.

Referring to FIG. 6, the memory block BLK_c includes a plurality of cell strings CS₁ to CS_m. The plurality of cell strings CS₁ to CS_m may be connected to a plurality of bit lines BL₁ to BL_m, respectively. Each of the cell strings CS₁ to CS_m includes at least one source select transistor SST, first to n-th memory cells MC₁ to MC_n, and at least one drain select transistor DST.

Each of the select transistors SST and DST and the memory cells MC₁ to MC_n may have a similar structure. In an embodiment, each of the select transistors SST and DST and the memory cells MC₁ to MC_n may include a channel layer, a tunneling insulating layer, a charge storage layer, and a blocking insulating layer. The source select transistor SST of each cell string is connected between a common source line CSL and the memory cells MC₁ to MC_n.

The first to n-th memory cells MC₁ to MC_n of each cell string are connected between the source select transistor SST and the drain select transistor DST.

The drain select transistor DST of each cell string is connected between a corresponding bit line and the memory cells MC₁ to MC_n.

Memory cells connected to the same word line configure one page. The cell strings CS₁ to CS_m may be selected by selecting the drain select line DSL. One page among the selected cell strings may be selected by selecting one of the word lines WL₁ to WL_n.

In another embodiment, even bit lines and odd bit lines may be provided instead of the first to m-th bit lines BL₁ to BL_m. Even-numbered cell strings among the cell strings CS₁ to CS_m may be connected to even bit lines, and odd-numbered cell strings may be connected to odd bit lines, respectively.

FIG. 7 is a diagram illustrating a storage area included in the storage device according to an embodiment of the present disclosure.

The semiconductor memory device 100 of the storage device 1000 includes the memory cell array 110, and data may be stored in the plurality of memory cells included in

15

the memory cell array **110**. Accordingly, the memory cell array **110** of the semiconductor memory device **100** may configure the storage area of the storage device **1000**. The storage area may be divided into a mapped area and an unmapped area. User data received from the host device **300** may be stored in the mapped area. A position of the user data stored in the storage device **1000** may be mapped by an address of data. That is, all data stored in the mapped area may be valid data.

Furthermore, from a viewpoint of the host device **300**, the unmapped area corresponds to an empty storage area. A file system of the host device **300** may recognize all unmapped areas as an empty space of the storage device **1000**. However, data may be stored in the unmapped area inside the storage device. Hereinafter, the unmapped area is described in more detail with reference to FIG. **8**.

FIG. **8** is a diagram illustrating the unmapped area of the storage area included in the storage device according to an embodiment of the present disclosure.

Referring to FIG. **8**, the unmapped area may include an invalid data area and a free area. The invalid data area may be an area in which invalid data is stored, and the free area may be an area including only memory cells of an erasure state.

Data stored in the mapped area is valid data, and when address mapping of the valid data is unmapped, the data becomes the invalid data. The invalid data is data that is no longer necessary from a standpoint of the host device **300**. The invalid data may be deleted by the erase operation of the storage device. When the invalid data stored in the invalid data area is deleted by the erase operation, an area corresponding to the invalid data may be changed to the free area.

FIG. **9** is a diagram illustrating the test operation of the semiconductor memory device in response to the request from the host device according to an embodiment of the present disclosure.

Referring to FIG. **9**, the host device **300** transmits a test request RQ_{TST} to the controller **200** of the storage device **1000** (①). The host device **300** may determine whether the storage device performs the test operation as necessary. In this case, the test request generator **310** of the host device **300** may generate the test request RQ_{TST} .

The test request RQ_{TST} may include information on a target of the test operation to be performed by the storage device. For example, the host device **300** may determine to perform the test operation only on the mapped area of the storage area included in the storage device **1000**. In this case, the test request RQ_{TST} transmitted from the host device **300** to the storage device **1000** may include information indicating that the target of the test operation is the mapped area. As another example, the host device **300** may determine to perform the test operation only on the unmapped area of the storage area included in the storage device **1000**. In this case, the test request RQ_{TST} transmitted from the host device **300** to the storage device **1000** may include information indicating that the target of the test operation is the unmapped area.

The test request RQ_{TST} may include information on the type of test operation to be performed by the storage device. For example, the host device **300** may determine that the storage device **1000** performs an entire test operation on a designated area. The “entire test” operation may refer to a test performed in a method of reading all data stored in the designated area. In this case, the test request RQ_{TST} transmitted from the host device **300** to the storage device **1000** may include information indicating that the type of test operation is the entire test. As another example, the host

16

device **300** may determine that the storage device **1000** performs a partial test operation on the designated area. The “partial test” operation may refer to a test performed in a method of reading some data among the data stored in the designated area. In this case, the test request RQ_{TST} transmitted from the host device **300** to the storage device **1000** may include information indicating that the type of the test operation is the partial test.

The memory test controller **210** of the controller **200** may perform a read test operation on the semiconductor memory device **100** based on the received test request RQ_{TST} . Specifically, the memory test controller **210** of the controller **200** may generate the read command CMD_{RD} corresponding to the received test request RQ_{TST} and transmit the generated read command CMD_{RD} to the semiconductor memory device **100** (②). According to an embodiment, the memory test controller **210** of the controller **200** may generate a plurality of read commands and transmit the plurality of generated read commands to the semiconductor memory device **100**.

The semiconductor memory device **100** may perform the read operation in response to the received read command CMD_{RD} . The semiconductor memory device **100** may transmit the data $DATA$ read as a result of the read operation to the controller **200** (③).

The controller **200** may analyze the read data $DATA$ (④). In an embodiment, the controller **200** may perform an error correction operation on the read data $DATA$. When the read data $DATA$ includes few error bits, the error correction operation may be passed on the read data $DATA$. When the read data $DATA$ includes many error bits, at least one error correction operation on the read data $DATA$ may be failed.

The controller **200** may transmit test result information TRI including an analysis result of the read data to the host device **300** (⑤). For example, the test result information TRI may include the number of error bits included in the read data $DATA$. As another example, the test result information TRI may include the number of failures of the error correction operation occurring on the read data $DATA$.

FIG. **10** is a diagram illustrating the test request received from the host device according to an embodiment of the present disclosure.

Referring to FIG. **10**, the test request RQ_{TST} received from the host device **300** to the storage device **1000** may include two data fields. A first data field included in the test request RQ_{TST} may include information regarding the type of the test to be performed by the storage device **1000**. For example, the first data field of the test request RQ_{TST} may include data indicating whether the test to be performed by the storage device **1000** is the “entire test” or the “partial test”. Moreover, a second data field included in the test request RQ_{TST} may include information regarding an area indicated by the target of the test operation to be performed by the storage device **1000**.

FIGS. **11A** and **11B** are diagrams illustrating an operation of the storage device that performs the entire test on the mapped area based on the test request from the host device according to an embodiment of the present disclosure.

Referring to FIG. **11A**, the test request RQ_{TST} received from the host device **300** is shown. Referring to the first data field of the test request RQ_{TST} of FIG. **11A**, the test operation to be performed may be the “entire test” operation. Furthermore, referring to the second data field of the test request RQ_{TST} of FIG. **11A**, the target of the test operation to be performed may be the mapped area.

Referring to FIG. **11B**, the target of the test operation to be performed by the storage device **1000** in response to the

17

test request RQ_{TST} according to FIG. 11A is shown. Since the test operation to be performed is the “entire test” operation, and the target of the test operation to be performed is the mapped area, the entire data stored in the mapped area becomes the test target as shown in FIG. 11B. In this case, the storage device 1000 may read all data stored in the mapped area and analyze the read data. For example, the storage device may read all data stored in the mapped area and perform the error correction operation on the read data. In an embodiment, the storage device may transmit the number of error bits, an error bit rate, or the like generated as a result of the error correction operation to the host device 300 as the test operation result information TRI. In another embodiment, the storage device may count the number of failures of error correction occurring as a result of the error correction operation, and transmit a count value to the host device 300 as the test operation result information TRI.

FIGS. 12A and 12B are diagrams illustrating an operation of the storage device that performs the partial test on the mapped area based on the test request from the host device according to an embodiment of the present disclosure.

Referring to FIG. 12A, the test request RQ_{TST} received from the host device 300 is shown. Referring to the first data field of the test request RQ_{TST} of FIG. 12A, the test operation to be performed may be the “partial test” operation. Referring to the second data field of the test request RQ_{TST} of FIG. 12A, the target of the test operation to be performed may be the mapped area.

Referring to FIG. 12B, the target of the test operation to be performed by the storage device 1000 in response to the test request RQ_{TST} according to FIG. 12A is shown. Since the test operation to be performed is the “partial test” operation and the target of the test operation to be performed is the mapped area, some data among the data stored in the mapped area become the test target as shown in FIG. 12B.

The storage device 1000 may sample some of the data stored in the mapped area and determine the sampled some data as the test target. In FIG. 12B, a test target 1, a test target 2, and a test target 3 are selected.

The storage device 1000 may read data corresponding to the selected test targets 1 to 3 and analyze the read data. For example, the storage device may read data corresponding to the test targets 1 to 3 stored in the mapped area and perform the error correction operation on the read data. In an embodiment, the storage device may transmit the number of error bits, the error bit rate, or the like generated as a result of the error correction operation to the host device 300 as the test operation result information TRI. In another embodiment, the storage device may count the number of error correction failures occurring as a result of the error correction operation, and transmit the count value to the host device 300 as the test operation result information TRI.

In a case of the partial test operation, a test performance time is shorter than that of the entire test operation. Therefore, in this case, the storage device 1000 may quickly generate the test result information TRI and transmit the test result information TRI to the host device 300. On the other hand, when the entire test operation is performed, the test result information TRI which is more accurate than that of the partial test operation, may be generated.

FIGS. 13A and 13B are diagrams illustrating an operation of the storage device that performs the entire test on the unmapped area based on the test request from the host device according to an embodiment of the present disclosure.

Referring to FIG. 13A, the test request RQ_{TST} received from the host device 300 is shown. Referring to the first data

18

field of the test request RQ_{TST} of FIG. 13A, the test operation to be performed may be the “entire test” operation. Referring to the second data field of the test request RQ_{TST} of FIG. 13A, the target of the test operation to be performed may be the unmapped area.

Referring to FIG. 13B, the target of the test operation to be performed by the storage device 1000 in response to the test request RQ_{TST} according to FIG. 13A is shown. Since the test operation to be performed is the “entire test” operation and the target of the test operation to be performed is the unmapped area, entire data stored in the invalid data area of the unmapped area as shown in FIG. 13B become the test target. Since data is not stored in the free area, the free area does not become the test target.

The storage device 1000 may determine the entire data stored in the invalid data area of the unmapped area as the test target. The storage device 1000 may read the entire invalid data stored in the invalid data area and analyze the read data. For example, the storage device may read the entire invalid data stored in the invalid data area and perform the error correction operation on the read data. In an embodiment, the storage device may transmit the number of error bits, the error bit rate, or the like generated as a result of the error correction operation to the host device 300 as the test operation result information TRI. In another embodiment, the storage device may count the number of error correction failures occurring as a result of the error correction operation, and transmit the count value to the host device 300 as the test operation result information TRI.

FIGS. 14A and 14B are diagrams illustrating an operation of the storage device that performs the partial test on the unmapped area based on the test request from the host device according to an embodiment of the present disclosure.

Referring to FIG. 14A, the test request RQ_{TST} received from the host device 300 is shown. Referring to the first data field of the test request RQ_{TST} of FIG. 14A, the test operation to be performed may be the “partial test” operation. Referring to the second data field of the test request RQ_{TST} of FIG. 14A, the target of the test operation to be performed may be the unmapped area.

Referring to FIG. 14B, the target of the test operation to be performed by the storage device 1000 in response to the test request RQ_{TST} according to FIG. 14A is shown. Since the test operation to be performed is the “partial test” operation, and the target of the test operation to be performed is the unmapped area, some data among the invalid data stored in the invalid data area of the unmapped area as shown in FIG. 14B become the test target.

The storage device 1000 may sample some of the invalid data stored in the invalid data area and determine the sampled some data as the test target. In FIG. 14B, a test target 1 and a test target 2 are selected.

The storage device 1000 may read data corresponding to the selected test targets 1 and 2 and analyze the read data.

For example, the storage device may read invalid data corresponding to the test targets 1 and 2 stored in the invalid data area and perform the error correction operation on the read invalid data. In an embodiment, the storage device may transmit the number of error bits, the error bit rate, or the like generated as a result of the error correction operation to the host device 300 as the test operation result information TRI. In another embodiment, the storage device may count the number of error correction failures occurring as a result of the error correction operation, and transmit the count value to the host device 300 as the test operation result information TRI.

FIG. 15 is a flowchart illustrating a method of operating a host device according to an embodiment of the present disclosure.

Referring to FIG. 15, the method of operating the host device 300 according to an embodiment of the present disclosure includes determining the area to be tested among the mapped area and the unmapped area (S110), determining the type of the test to be performed on the determined area (S130), generating the test request corresponding to the determined area and the type (S150), and transmitting the generated test request to the storage device (S170).

In operation S110, the test request generator 310 of the host device 300 may determine the area to be tested in the storage area of the storage device. As described above with reference to FIGS. 11A to 14B, the host device 300 may determine one of the mapped area and the unmapped area as the area to be tested.

In operation S130, the test request generator 310 of the host device 300 may determine the type of the test to be performed on the area determined in operation S110. As described above, the test operation may be one of the “entire test” operation and the “partial test” operation.

In operation S150, the test request generator 310 of the host device 300 generates the test request corresponding to the determined area and the type. As described above with reference to FIGS. 10 to 14B, the test request RQ_{TST} may include the data field indicating the type and the data field indicating the test target area.

In operation S170, the test request generator 310 of the host device 300 transmits the generated test request to the storage device. The storage device 1000 may perform the test operation in response to the received test request. Hereinafter, a test operation of a storage device according to an embodiment of the present disclosure is described in more detail with reference to FIGS. 16 and 17.

FIG. 16 is a flowchart illustrating a method of operating a storage device according to an embodiment of the present disclosure.

Referring to FIG. 16, the method of operating the storage device according to an embodiment of the present disclosure includes receiving the test request from the host device 300 (S210), determining the data area on which the test operation is performed based on the test request (S230), controlling the semiconductor memory device to read the data stored in the determined data area (S250), analyzing the read data to generate the test result information (S270), and transmitting the generated test result information to the host device (S290).

In operation S210, the controller 200 of the storage device 1000 receives the test request RQ_{TST} from the host device 300, and in operation S230, the controller 200 determines the data area on which the test operation is performed based on the test request RQ_{TST} . For example, as shown in FIG. 11A, when the first data field of the test request RQ_{TST} indicates the “entire test” operation and the second data field indicates the mapped area, the controller 200 may determine the entire mapped area as the data area on which the test operation is performed as shown in FIG. 11B. As another example, as shown in FIG. 14A, when the first data field of the test request RQ_{TST} indicates the “partial test” operation and the second data field indicates the unmapped area, the controller 200 may determine some of the invalid data area in which the invalid data is stored in the unmapped area as the data area on which the test operation is performed as shown in FIG. 14B.

In operation S250, the controller 200 may control the semiconductor memory device to read the data stored in the

determined data area. As an example, in operation S250, the controller 200 may generate at least one read command corresponding to the determined data area and transmit the generated read command to the semiconductor memory device. Operation S250 is described in more detail with reference to FIG. 17.

According to operation S250, the semiconductor memory device 100 may read the data stored in the determined data area, and transmit the read data to the controller 200. In operation S270, the controller 200 may analyze the read data to generate test result information TRI. As described above, in operation S250, the controller may perform the error correction operation on the read data. A result of performing the error correction operation may be included in the test result information TRI.

In operation S290, the controller 200 may transmit the test result information TRI to the host device 300. Accordingly, the host device 300 may utilize the test result information TRI to back up data stored in a degraded data area.

FIG. 17 is a flowchart illustrating operation S250 of FIG. 16 according to an embodiment of the present disclosure.

Referring to FIG. 17, operation S250 of FIG. 16 includes generating a command for reading unit data among the data stored in the determined data area (S310), transmitting the generated command to the semiconductor memory device (S330), receiving the data from the semiconductor memory device (S350), performing the error correction operation on the received data (S370), and determining whether all data stored in the determined data area are received (S390).

A plurality of pieces of unit data may be stored in the data area determined as the test target. As an example, the unit data may be page data. In operation S310, the command for reading the unit data among the data stored in the data area determined as the test target may be generated. The controller 200 may transmit the command generated in operation S310 to the semiconductor memory device. The semiconductor memory device may read corresponding unit data in response to the command, and transmit the read unit data to the controller.

In operation S350, the controller 200 receives the unit data from the semiconductor memory device. In operation S370, the controller 200 may perform the error correction operation on the received unit data. In operation S390, the controller 200 determines whether all data stored in the determined data area are received. When all data stored in the determined data area are received (S390: Yes), operation S250 is ended. When the all data stored in the determined data area are not received (S390: No), the process returns to operation S310 to generate a command for reading another unit data. Thereafter, steps S330, S350, S370, and S390 may be repeatedly performed.

FIG. 18 is a block diagram illustrating a storage device including a semiconductor memory device and a controller according to an embodiment of the present disclosure.

The semiconductor memory device 1300 of FIG. 18 may be configured and may operate similarly to the semiconductor memory device 100 described with reference to FIG. 2.

The controller 1200 is connected to a host device Host and the semiconductor memory device 1300. The controller 1200 is configured to access the semiconductor memory device 1300 in response to a request from the host device Host. For example, the controller 1200 is configured to control read, program, erase, and background operations of the semiconductor memory device 1300. The controller 1200 is configured to provide an interface between the semiconductor memory device 1300 and the host device

21

Host. The controller **1200** is configured to drive firmware for controlling the semiconductor memory device **1300**.

The controller **1200** includes a random access memory (RAM) **1210**, a processing unit **1220**, a host interface **1230**, a memory interface **1240**, and an error correction block **1250**.

The RAM **1210** is used as one of an operation memory of the processing unit **1220**, a cache memory between the semiconductor memory device **1300** and the host device Host, and a buffer memory between the semiconductor memory device **1300** and the host device Host.

The processing unit **1220** controls an overall operation of the controller **1200**. The processing unit **1220** is configured to control the read program, erase, and background operations of the semiconductor memory device **1300**. The processing unit **1220** is configured to drive firmware for controlling the semiconductor memory device **1300**. The processing unit **1220** may perform a function of a flash translation layer (FTL). The processing unit **1220** may convert a logical block address (LBA) provided by the host device into a physical block address (PBA) through the FTL. The FTL may receive the logical block address (LBA) and convert the LBA into the PBA by using a mapping table. There are several address mapping methods of the FTL according to a mapping unit. A representative address mapping method includes a page mapping method, a block mapping method, and a hybrid mapping method.

The host interface **1230** includes a protocol for performing data exchange between the host device Host and the controller **1200**. In an embodiment, the controller **1200** is configured to communicate with the host device Host through at least one of various communication standards or interfaces such as a universal serial bus (USB) protocol, a multimedia card (MMC) protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-e or PCIe) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer system interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, and a private protocol.

The memory interface **1240** interfaces with the semiconductor memory device **1300**. For example, the memory interface **1240** includes a NAND interface or a NOR interface.

The error correction block **1250** is configured to detect and correct an error of data received from the semiconductor memory device **1300** using an error correcting code (ECC). The error correction block **1250** may correct an error by using the ECC on read page data. The error correction block **1250** may correct an error by using a coded modulation such as a low density parity check (LDPC) code, a Bose, Chaudhri, Hocquenghem (BCH) code, a turbo code, a Reed-Solomon code, a convolution code, a recursive systematic code (RSC), a trellis-coded modulation (TCM), a block coded modulation (BCM), and a hamming code.

During a read operation, the error correction block **1250** may correct an error of the read page data. Decoding may be failed when the read page data includes error bits that exceed a correctable number of bits. The decoding may be successful when the page data includes error bits equal to or less than the correctable number of bits. The success of the decoding indicates that a read command is passed. The failure of the decoding indicates that the read command is failed. When the decoding is successful, the controller **1200** outputs the page data in which the error is corrected to the host.

22

The controller **200** of FIG. **1** may be implemented as the controller **1200** shown in FIG. **18**. In this case, the memory test controller **210** of FIG. **1** may be implemented as firmware driven by the processing unit **1220**.

The controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device. In an embodiment, the controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device to configure a memory card. For example, the controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device to configure a memory card such as a PC card (personal computer memory card international association (PCMCIA)), a compact flash (CF) card, a smart media card (SM or SMC), a memory stick, a multimedia card (MMC, RS-MMC, or MMCmicro), a secure digital (SD) card (SD, miniSD, microSD, or SDHC), and a universal flash storage (UFS).

The controller **1200** and the semiconductor memory device **1300** may be integrated into one semiconductor device to configure a semiconductor drive (solid state drive (SSD)). The semiconductor drive (SSD) includes a storage device configured to store data in the semiconductor memory. When the storage device is used as the semiconductor drive (SSD), an operation speed of the host device Host connected to the storage device is dramatically improved.

As another example, the storage device **1000** is provided as one of various components of an electronic device such as a computer, an ultra-mobile PC (UMPC), a workstation, a net-book, a personal digital assistants (PDA), a portable computer, a web tablet, a wireless phone, a mobile phone, a smart phone, an e-book, a portable multimedia player (PMP), a portable game machine, a navigation device, a black box, a digital camera, a 3-dimensional television, a digital audio recorder, a digital audio player, a digital picture recorder, a digital picture player, a digital video recorder, a digital video player, a device capable of transmitting and receiving information in a wireless environment, one of various electronic devices configuring a home network, one of various electronic devices configuring a computer network, one of various electronic devices configuring a telematics network, an RFID device, or one of various components configuring a computing system.

In an embodiment, the semiconductor memory device **1300** or the storage device may be mounted as a package of various types. For example, the semiconductor memory device **1300** or the storage device may be packaged and mounted in a method such as a package on package (POP), ball grid arrays (BGAs), chip scale packages (CSPs), plastic leaded chip carriers (PLCC), a plastic dual in line package (PDIP), a die in wafer pack, die in wafer form, a chip on board (COB), a ceramic dual in line package (CERDIP), a plastic metric quad flat pack (MQFP), a thin quad flat pack (TQFP), a small outline integrated circuit (SOIC) package, a shrink small outline package (SSOP), a thin small outline package (TSOP), a system in package (SIP), a multi-chip package (MCP), a wafer-level fabricated package (WFP), or a wafer-level processed stack package (WSP).

FIG. **19** is a block diagram illustrating an application example **2000** of the storage device of FIG. **18** according to an embodiment of the present disclosure.

Referring to FIG. **19**, the storage device **2000** includes a semiconductor memory device **2100** and a controller **2200**. The semiconductor memory device **2100** includes a plurality of semiconductor memory chips. The plurality of semiconductor memory chips are divided into a plurality of groups.

23

In FIG. 19, the plurality of groups communicate with the controller 2200 through first to k-th channels CH1 to CHk, respectively. Each semiconductor memory chip is configured and is operated similarly to one of the semiconductor memory device 1300 described with reference to FIG. 18.

Each group is configured to communicate with the controller 2200 through one common channel. The controller 2200 is configured similarly to the controller 1200 described with reference to FIG. 18 and is configured to control the plurality of memory chips of the semiconductor memory device 2100 through the plurality of channels CH1 to CHk.

In FIG. 19, the plurality of semiconductor memory chips are connected to one channel. However, it will be understood that the storage device 2000 may be modified so that one semiconductor memory chip is connected to one channel.

FIG. 20 is a block diagram illustrating a computing system including the storage device described with reference to FIG. 19 according to an embodiment of the present disclosure.

Referring to FIG. 20, the computing system 3000 includes a central processing unit 3100, a random access memory (RAM) 3200, a user interface 3300, a power supply 3400, a system bus 3500, and the storage device 2000.

The storage device 2000 is electrically connected to the central processing unit 3100, the RAM 3200, the user interface 3300, and the power supply 3400 through the system bus 3500. Data provided through the user interface 3300 or processed by the central processing unit 3100 is stored in the storage device 2000.

In FIG. 20, the semiconductor memory chip 2100 is connected to the system bus 3500 through the controller 2200. However, the semiconductor memory chip 2100 may be configured to be directly connected to the system bus 3500. At this time, a function of the controller 2200 is performed by the central processing unit 3100 and the RAM 3200.

In FIG. 20, the storage device 2000 described with reference to FIG. 19 is provided. However, the storage device 2000 may be replaced with the storage device 1000 described with reference to FIG. 18. In an embodiment, the computing system 3000 may be configured to include both of the storage devices 1000 and 2000 described with reference to FIGS. 18 and 19.

The embodiments of the present disclosure disclosed in the present specification and drawings are merely provided with specific examples to easily describe the technical content of the present disclosure and to help understanding of the present disclosure, and are not intended to limit the scope of the present disclosure. It will be apparent to those of ordinary skill in the art that other modified examples based on the technical spirit of the present disclosure may be implemented in addition to the embodiments disclosed herein and the following claims. Furthermore, the embodiments may be combined to form additional embodiments.

What is claimed is:

1. A controller comprising:

a memory test controller configured to:

receive a test request which indicates a position of a test area and a range of the test area from a host device, and control a memory device to perform a read operation on the test area; and

a buffer configured to store read data for the read operation,

wherein the memory test controller determines the test area among a mapped area or an unmapped area

24

included in the memory device, generates test information of the read data and transmits the test information to the host device,

wherein the mapped area stores valid data to which an address is mapped, and

wherein the unmapped area stores invalid data from which the address of the valid data is unmapped.

2. The controller of claim 1, wherein the unmapped area includes an invalid data area and a free area, and the invalid data is stored in the invalid data area.

3. The controller of claim 1, wherein the test request includes a first field indicating the test area on which the read operation is performed and a second field indicating a type of the read operation to be performed by the memory device, and

wherein the type of the read operation includes an entire read and a partial read.

4. The controller of claim 1, wherein the memory test controller is configured to:

transmit a command for reading unit data among the read data stored in the test area;

store the read data received from the memory device in the buffer;

perform an error correction operation on the read data; and

generate a test result information based on a result of the error correction operation.

5. The controller of claim 4, wherein the test result information includes information on at least one of a number of error bits in the unit data read from the test area and a number of times that an error correction fails during the error correction operation.

6. The controller of claim 4, wherein the unit data is page data.

7. A method of a controller, the method comprising:

receiving a test request which indicates a position of a test area and a range of the test area from a host device;

determining the test area among a mapped area or an unmapped area included in the memory device;

controlling a memory device to perform a read operation on the test area;

generating test information of read data of the read operation; and

transmitting the test information to the host device,

wherein the mapped area stores valid data to which an address is mapped, and

wherein the unmapped area stores invalid data from which the address of the valid data is unmapped.

8. The method of claim 7, wherein the unmapped area includes an invalid data area and a free area, and the invalid data is stored in the invalid data area.

9. The method of claim 7, wherein the test request includes a first field indicating the test area on which the read operation is performed and a second field indicating a type of the read operation to be performed by the memory device, and

wherein the type of the read operation includes an entire read and a partial read.

10. The method of claim 7, wherein the controlling the memory device to perform the read operation comprises:

transmitting a command for reading unit data among the read data stored in the test area; and

receiving the read data from the memory device.

11. The method of claim 10, wherein generating the test information comprises:

performing an error correction operation on the read data; and

25

generating a test result information based on a result of the error correction operation.

12. The method of claim **11**, wherein the test result information includes information on at least one of a number of error bits in the unit data read from the test area and a number of times that an error correction fails during the error correction operation. 5

13. The method of claim **11**, wherein the unit data is page data.

* * * * *

10

26